PROJECT REPORT

Data-Driven Social Engagement Initiative

A project using Python, SQL, sentiment analysis, and Power BI

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Abstract

The Data-Driven Social Engagement Initiative examines social media post performance using a combined workflow of Python, SQL, sentiment analysis, and Power BI. The post-level dataset contains 29,999 records and 23 columns, covering post attributes, publishing time, engagement metrics, and performance labels. A separate comment dataset was analysed to explore sentiment and explicit patterns of relatability. Python was used for data validation, feature engineering, descriptive analysis, and a Random Forest classification experiment. SQL was used to investigate category performance, highly engaging posts, saves, media types, and monthly trends. Power BI was used to build an interactive dashboard with KPI cards, category and media comparisons, a monthly engagement trend, and slicers for category, media type, and date. The results show that average engagement rates across categories and media types are close, and tested comparisons did not establish statistically significant differences. The classifier achieved 23.88% accuracy, close to the approximate 25% chance level for four balanced classes. Comment sentiment was 46.2% neutral, 34.1% positive, and 19.7% negative among 9,960 cleaned comments. The report emphasizes cautious interpretation: the findings are observational, the relatability detector is conservative, and the model should not be treated as a reliable predictor without further work.

Table of Contents

- 1. Introduction
- 2. Problem Statement, Aim and Objectives
- 3. Dataset Description
- 4. Tools and Technologies
- 5. Methodology
- 6. Implementation
- 7. Results and Discussion
- 8. Recommendations
- 9. Limitations
- 10. Conclusion and Future Scope
- 11. References
- Appendix A. SQL Query Examples
- Appendix B. Dashboard Screenshot Placeholders

1. Introduction

Social media platforms generate large volumes of data about content, audience response, and publishing activity. Likes, comments, shares, saves, reach, impressions, and follower gains can provide different views of how audiences interact with posts. However, these measures need to be organised and interpreted carefully: a high total may reflect a larger number of posts, while a high average rate may indicate stronger per-post performance.

This project applies a practical analytics workflow to social media post and comment data. Python supports cleaning, feature creation, and exploratory analysis; SQL provides repeatable queries for aggregation and ranking; sentiment analysis summarizes comment tone; and Power BI presents key metrics and filters in an interactive dashboard.

2. Problem Statement, Aim and Objectives

2.1 Problem statement

The project investigates how post attributes and publishing context relate to engagement outcomes, which categories and individual posts have comparatively high observed engagement, and what comment sentiment can reveal about audience response. The analysis also evaluates whether the available features can predict the provided performance bucket label.

2.2 Aim

To analyse social media engagement data using Python, SQL, sentiment analysis, and Power BI, and to communicate the findings through a reproducible analytical workflow and an interactive dashboard.

2.3 Objectives

- Validate the post dataset for missing values, duplicate records, identifier uniqueness, and date range.
- Create engagement-related features such as like rate, share rate, save rate, comment rate, and virality score.
- Use SQL aggregation, ranking, window functions, and date-based queries to compare posts and categories.
- Explore the polarity of cleaned comments and test a conservative pattern-based relatability detector.
- Evaluate a Random Forest classifier using pre-publication features and a held-out test set.
- Develop a Power BI dashboard with KPI cards, comparison charts, a monthly trend, and interactive slicers.
- Translate findings into cautious, evidence-based recommendations and document limitations.

3. Dataset Description

3.1 Post-level dataset

The main dataset contains 29,999 rows and 23 columns. The recorded post-date range is 19 November 2024 to 19 November 2025. Initial validation reported zero missing values and zero duplicate rows; post_id was unique across all 29,999 records.

Field group	Example columns	Purpose
Identifiers and account	post_id, account_id, account_type, follower_count	Identify posts and describe account context

Content attributes	media_type, content_category, traffic_source, has_call_to_action	Compare content formats and attributes
Publishing context	post_datetime, post_date, post_hour, day_of_week	Explore timing-related patterns
Engagement outcomes	likes, comments, shares, saves, reach, impressions, engagement_rate, followers_gained	Measure observed audience response
Text metadata and label	caption_length, hashtags_count, performance_bucket_label	Describe captions and evaluate the supplied performance label

3.2 Comment dataset

A separate comment dataset initially contained 14,207 rows with the fields text and username. After cleaning, 9,960 comment rows were used for sentiment analysis. The exact cleaning rules and source URL should be documented in the final notebook or added here.

3.3 Data quality checks

- Post count: 29,999.
- Column count: 23.
- Missing values: 0 reported in the main dataset.
- Duplicate rows: 0 reported in the main dataset.
- Unique post IDs: 29,999.
- Date range: 2024-11-19 to 2025-11-19.

4. Tools and Technologies

Tool / technique	Role in the project
Python	Data processing, feature engineering, exploratory analysis, statistical testing, and model evaluation
Pandas and NumPy	Tabular data handling and numerical operations
Matplotlib / Seaborn	Exploratory charts and comparisons
VADER sentiment analysis	Lexicon-based comment sentiment classification
Scikit-learn Random Forest	Classification experiment for performance_bucket_label
MySQL	Aggregation, ranking, subqueries, window functions, and time-based queries
Power BI Desktop	Interactive KPI dashboard, charts, and slicers

5. Methodology

5.1 Data loading and validation

The post dataset was loaded into Python and MySQL. Row and column counts, missing values, duplicate rows, identifier uniqueness, and the date range were checked before analysis.

5.2 Feature engineering

Rates were derived by dividing engagement actions by reach. The virality score combined share rate, save rate, and comment rate with weights of 0.4, 0.4, and 0.2 respectively. Rows with zero reach should be checked before division; the exact handling should be documented from the notebook.

5.3 Exploratory analysis

Category, media type, publishing context, and engagement outcomes were compared using averages, totals, charts, and statistical tests. Totals and averages were interpreted separately because they answer different questions.

5.4 SQL analysis

SQL queries aggregated engagement by category, ranked top posts within categories and media types, identified high-save posts, compared category averages with the overall average, and summarized monthly engagement.

5.5 Sentiment analysis

VADER was applied to the cleaned comment text. The resulting labels were summarized as positive, neutral, and negative. A separate conservative pattern-based detector was used to identify explicit relatability expressions.

5.6 Predictive modelling

A Random Forest classifier used pre-publication features: follower_count, media_type, content_category, traffic_source, has_call_to_action, post_hour, day_of_week, caption_length, and hashtags_count. The data was split into training and test subsets, with 20% held out for testing and stratification used.

5.7 Dashboard development

Power BI was used to show five KPIs, average engagement by content category, average engagement by media type, a monthly engagement trend, and slicers for content category, media type, and date range.

6. Implementation

6.1 Python implementation

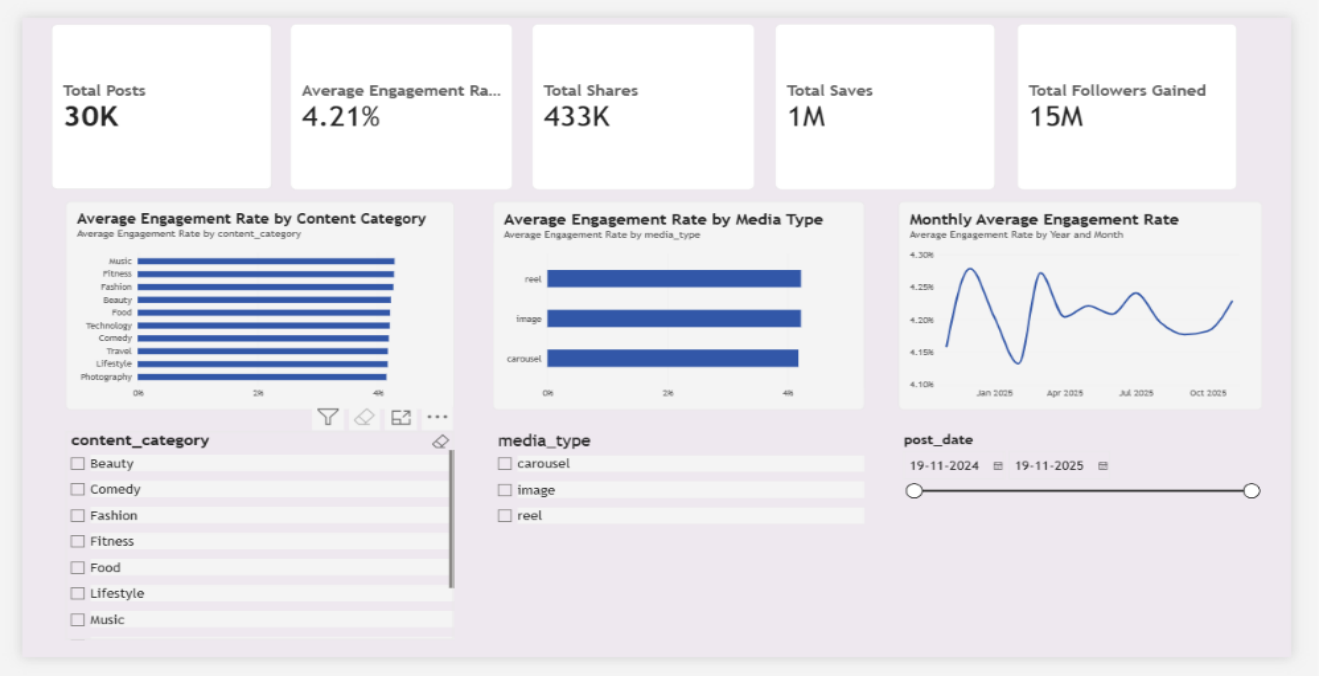
The Python workflow covered data validation, engineered engagement rates, exploratory comparisons, comment sentiment, and a supervised classification experiment. The derived metrics included share_rate, save_rate, comment_rate, like_rate, and virality_score. The mean virality_score was approximately 0.003897 and the maximum was approximately 0.026955 in the analysed data.

6.2 SQL implementation

The MySQL database was named social_engagement_db and the main table was posts. Queries used GROUP BY, aggregate functions, subqueries, DENSE_RANK() window functions, and date functions. The table was validated at 29,999 rows with unique post identifiers.

6.3 Power BI implementation

The report page contains five KPI cards: Total Posts, Average Engagement Rate, Total Shares, Total Saves, and Total Followers Gained. Three charts compare average engagement by content category, compare average engagement by media type, and show monthly average engagement. Three slicers allow filtering by content category, media type, and post date.



7. Results and Discussion

7.1 Category performance

Category	Posts	Average engagement	Total shares	Total saves
Music	3,003	4.28%	43,698	129,215
Fitness	3,004	4.27%	43,440	128,029
Fashion	3,034	4.26%	44,984	132,092
Beauty	2,953	4.22%	43,024	126,363
Food	3,010	4.21%	43,636	128,571
Technology	3,025	4.20%	41,813	124,201

Comedy	2,950	4.19%	42,805	127,185
Travel	2,968	4.17%	41,895	123,242
Lifestyle	3,017	4.16%	43,430	126,388
Photography	3,035	4.15%	44,059	130,190

Music had the highest observed average engagement rate at 4.28%, while Fashion had the highest total shares (44,984) and saves (132,092). These are descriptive differences; the statistical comparisons performed in the project did not establish significant category differences.

7.2 Media-type comparison

Media type	Average engagement rate
Reels	4.226641%
Images	4.225635%
Carousels	4.182856%

Average engagement was very similar for Reels and images, with carousels slightly lower in this dataset. The difference is small and should not be interpreted as evidence that one format causes better engagement.

7.3 Monthly engagement

Month	Year	Average engagement	Posts
November	2024	4.16%	1,004
December	2024	4.28%	2,532
January	2025	4.20%	2,587
February	2025	4.13%	2,322
March	2025	4.27%	2,499
April	2025	4.20%	2,509
May	2025	4.22%	2,579
June	2025	4.21%	2,494
July	2025	4.24%	2,535
August	2025	4.20%	2,509
September	2025	4.18%	2,320
October	2025	4.18%	2,532
November	2025	4.23%	1,577

Monthly averages ranged from 4.13% in February 2025 to 4.28% in December 2024. The pattern contains modest fluctuations rather than a sustained monotonic trend. The partial months at the beginning and end of the dataset may have fewer records because the observed range begins and ends mid-month.

7.4 High-performing individual posts

SQL ranking identified the top three engagement-rate posts within each media type and category. These rankings are useful for case review, but individual top posts may be outliers and should not be used alone to infer general format performance.

Example post	Category / type	Engagement rate	Shares	Saves
IG0005227	Food / Carousel	27.10%	51	246
IG0009304	Photography / Carousel	25.44%	57	246
IG0028356	Fashion / Carousel	24.21%	441	1,125
IG0017637	Beauty / Image	25.31%	24	108
IG0017987	Beauty / Image	25.27%	213	552
IG0017615	Comedy / Reel	26.58%	51	195

7.5 Comment sentiment and relatability

Sentiment label	Count	Share of cleaned comments
Neutral	4,602	46.2%
Positive	3,391	34.1%
Negative	1,967	19.7%
Total	9,960	100.0%

Neutral sentiment was the largest group, followed by positive and negative sentiment. The pattern-based relatability detector matched only 4 of 9,960 cleaned comments (0.04%). These matches comprised 3 relatable-neutral and 1 strong-positive comment, with no strong-negative matches. This very low match rate reflects a conservative explicit-pattern detector and must not be interpreted as the true prevalence of relatability. The detector likely misses implicit or differently worded expressions.

7.6 Predictive model evaluation

The Random Forest classifier achieved 23.88% accuracy on the held-out test set. The four performance classes were approximately balanced at around 7,500 posts each, so a uniform chance-level reference is approximately 25%. The observed accuracy is below that rough reference, indicating that the tested pre-publication features did not provide useful predictive performance in this experiment. The model should not be presented as a reliable forecasting system.

7.7 Statistical interpretation

The tested comparisons involving call-to-action presence, media type, posting hour, content category, traffic source, and day of week returned p-values above 0.05. Under the chosen tests and assumptions, the analysis did not find statistically significant differences. A non-significant result does not prove that groups are identical; it indicates that this analysis did not establish a difference at the selected threshold.

8. Recommendations

- Use the dashboard to monitor engagement rates and counts together; totals can be affected by how many posts are published.
- Review posts with high saves and shares qualitatively to understand their content, audience, and context before reusing a format.
- Treat observed category differences as hypotheses for further testing rather than definitive rankings.
- For future content decisions, run controlled or clearly documented A/B tests where possible and compare rates using consistent definitions.
- Improve the predictive model by reviewing label construction, adding relevant features, checking leakage and class balance, tuning models, and comparing against a simple baseline.
- Improve relatability analysis with a manually labelled sample, broader linguistic patterns, and validation metrics such as precision and recall.
- Continue tracking performance over time with the Power BI date, category, and media-type slicers.

9. Limitations

- The analysis is observational and cannot establish causation.
- The dataset's platform, collection method, and representativeness must be documented; results may not generalize to other accounts or platforms.
- The Random Forest accuracy of 23.88% is near or below the approximate 25% reference for four balanced classes, so the model has limited predictive value.
- The relatability detector was deliberately conservative and matched only 4 comments; this does not measure all forms of relatability.
- Sentiment tools may misclassify sarcasm, slang, context-dependent language, mixed sentiment, and multilingual comments.
- Small observed differences between categories and media types were not statistically significant in the tests conducted.
- Some high-performing individual posts may be outliers; their characteristics require qualitative review.

10. Conclusion and Future Scope

The project established an end-to-end workflow for analysing social media engagement through Python, MySQL, sentiment analysis, and Power BI. Data quality checks confirmed 29,999 unique post records over the observed date range. SQL and Python provided descriptive insights into category performance, post rankings, monthly patterns, and comment sentiment. The Power BI dashboard consolidates key metrics and allows interactive filtering. The most important conclusion is that descriptive differences should be interpreted cautiously: tested group comparisons were not statistically significant, and the classifier did not outperform a simple chance-level reference.

Future work can add a larger and more representative dataset, contextual text features, validated human-labelled sentiment and relatability samples, controlled experiments, and improved predictive baselines. The dashboard could also be extended with drill-through pages, confidence intervals, and automated refresh if the data pipeline permits.

11. References

- [1] Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Proceedings of the International AAAI Conference on Web and Social Media.
- [2] pandas documentation: https://pandas.pydata.org/docs/
- [3] NumPy documentation: https://numpy.org/doc/
- [4] scikit-learn documentation: https://scikit-learn.org/stable/
- [5] Microsoft Power BI documentation: https://learn.microsoft.com/power-bi/
- [6] MySQL documentation: https://dev.mysql.com/doc/

Appendix A. SQL Query Examples

A.1 Average engagement and total shares/saves by category

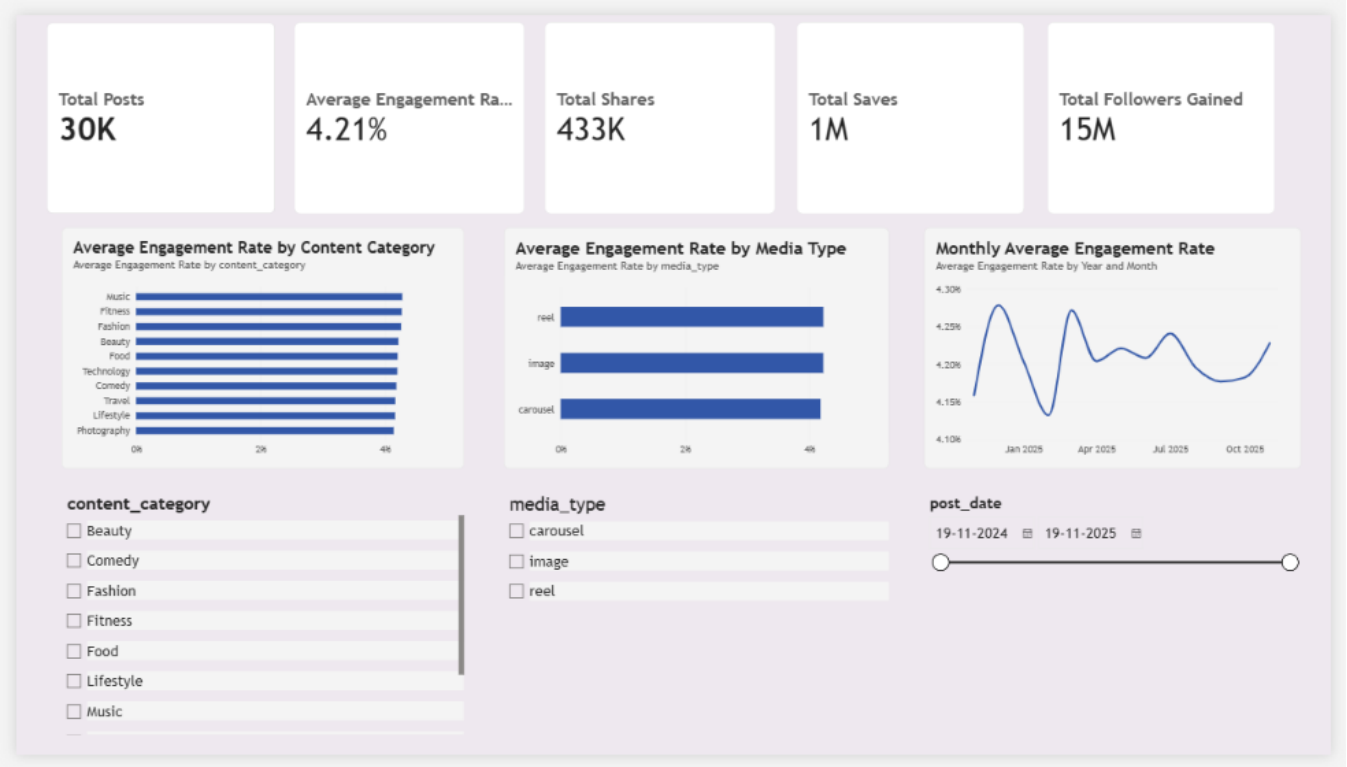
```
SELECT content_category,
       COUNT(*) AS total_posts,
       ROUND(AVG(engagement_rate) * 100, 2) AS avg_engagement_pct,
       SUM(shares) AS total_shares,
       SUM(saves) AS total_saves
FROM posts
GROUP BY content_category
ORDER BY avg_engagement_pct DESC;
```

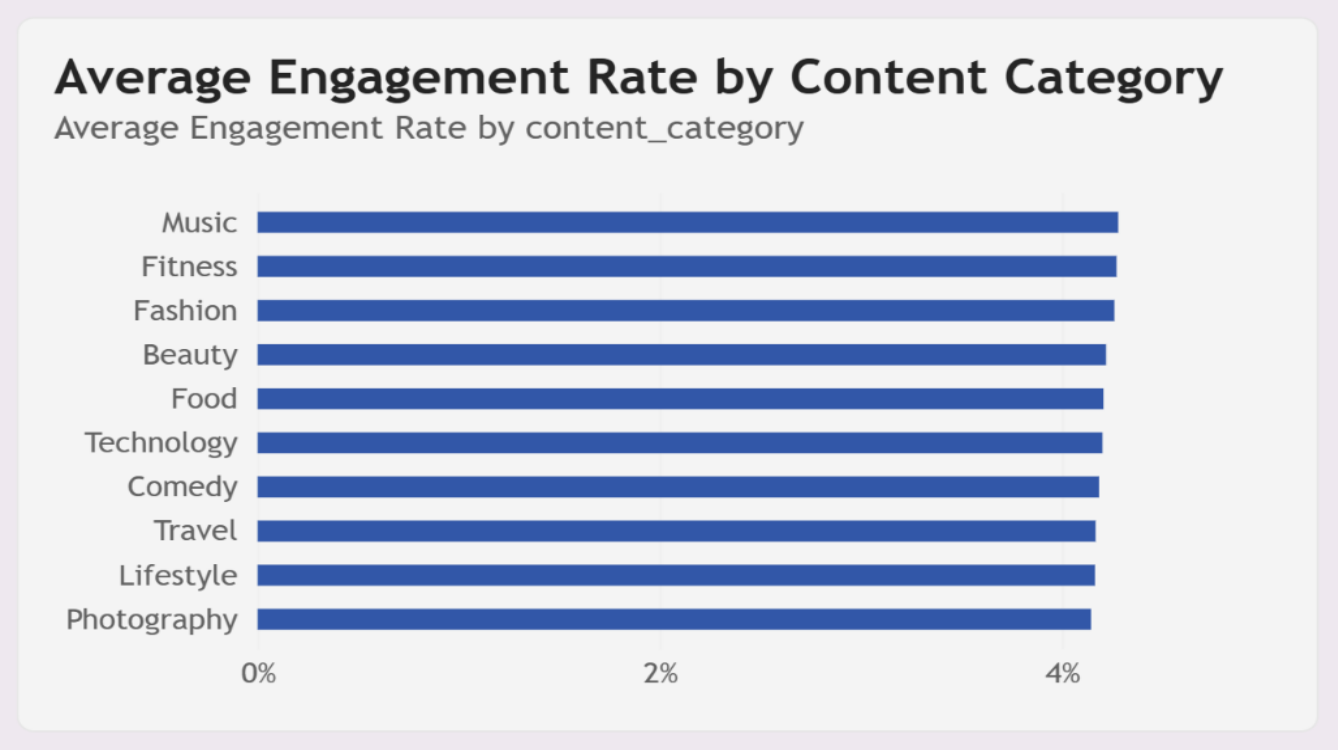
A.2 Example top posts by engagement within category (requires MySQL 8+)

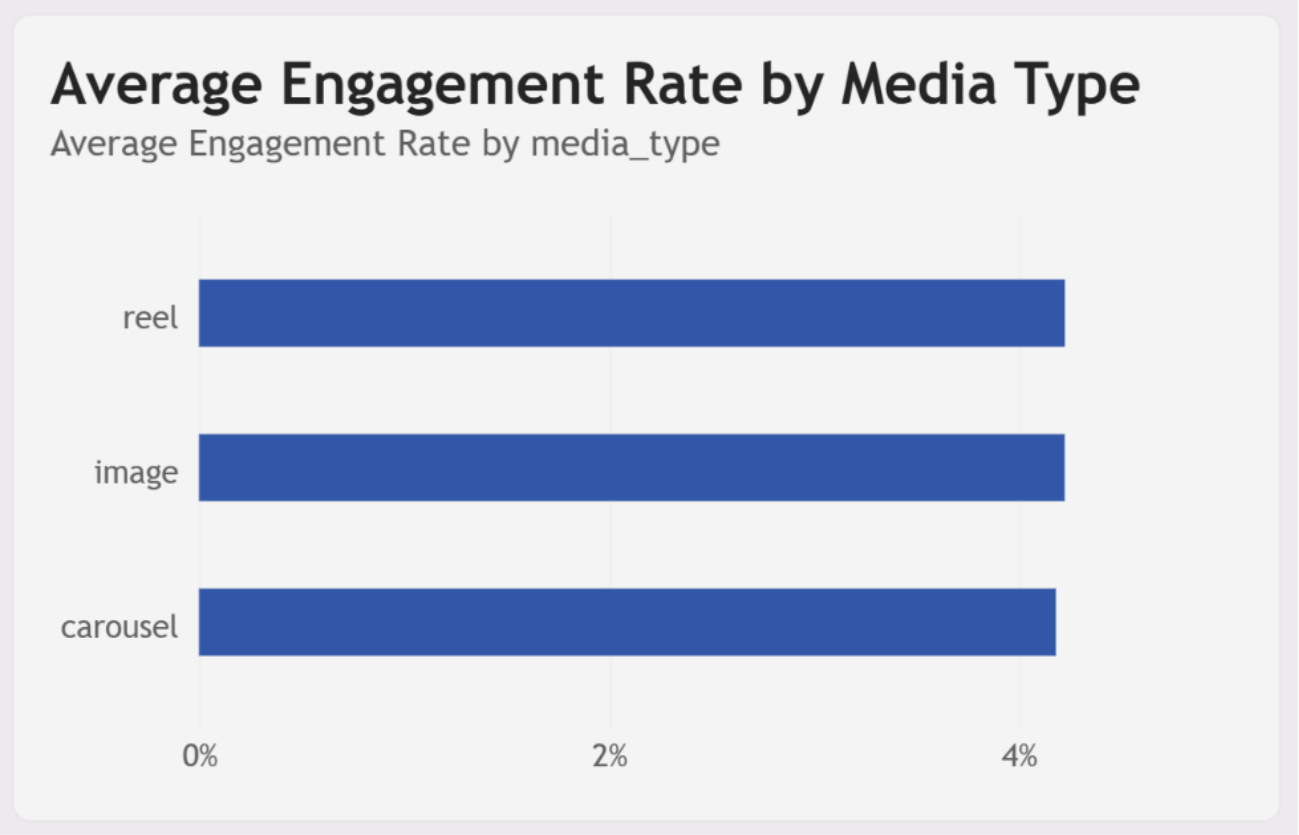
```
WITH ranked_posts AS (
    SELECT post_id, content_category, engagement_rate,
           DENSE_RANK() OVER (
               PARTITION BY content_category
               ORDER BY engagement_rate DESC
           ) AS engagement_rank
    FROM posts
)
SELECT *
FROM ranked_posts
WHERE engagement_rank <= 3
ORDER BY content_category, engagement_rank;
```

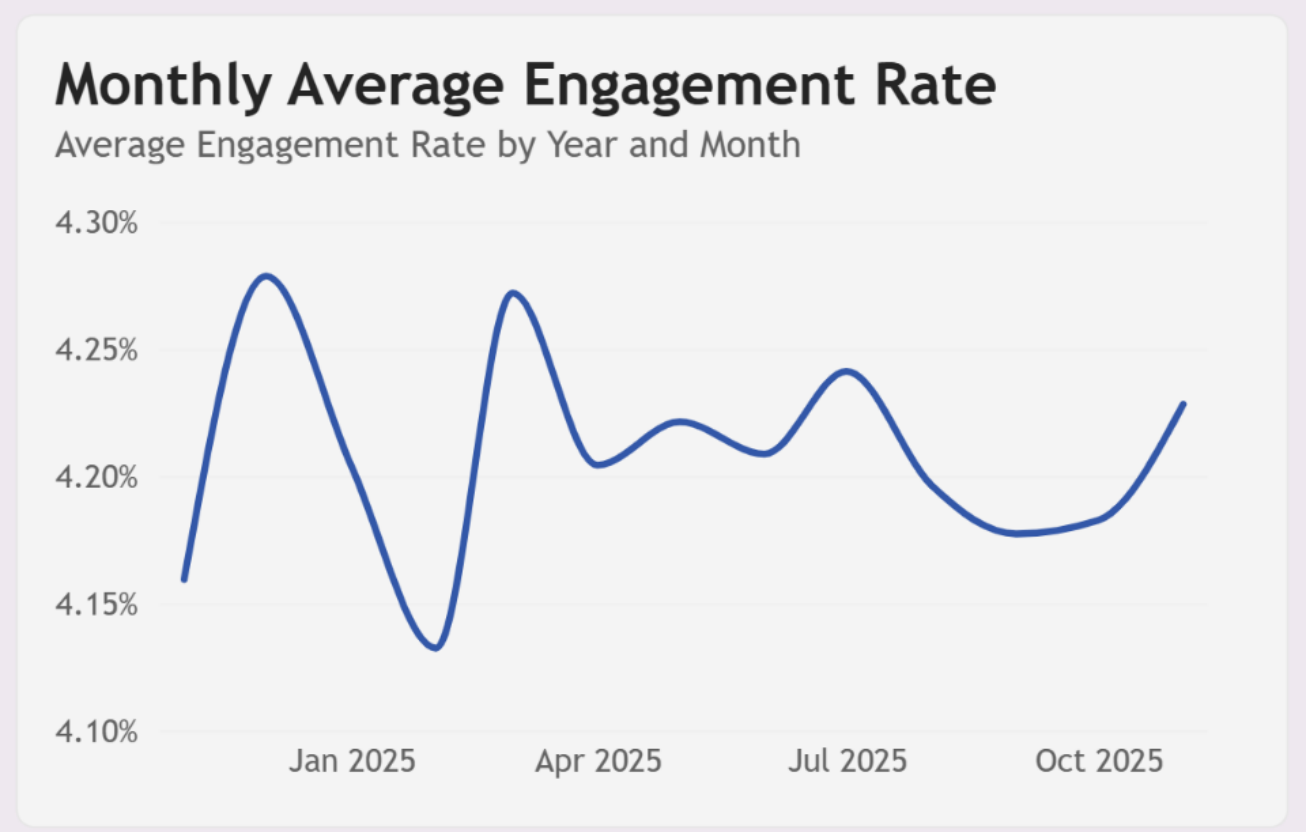
The full saved SQL script should be included as a separate deliverable or appended here if required.

Appendix B. Dashboard Screenshot Placeholders






Average Engagement Rate by Media Type
Average Engagement Rate by media_type
reel
image
carousel
0%
2%
4%


Monthly Average Engagement Rate
Average Engagement Rate by Year and Month
4.30%
4.25%
4.20%
4.15%
4.10%
Jan 2025
Apr 2025
Jul 2025
Oct 2025